

REST API Automation

restapi-automation-java-restassured-junit — Quick Reference

39

SCENARIOS

9

FEATURE FILES

9

RUNNERS

2

APIS TESTED

Architecture

Feature Files → Step Definitions → Services → Models / POJOs

```
src/test/java/com/automationexercise/api/  
  config/ ApiConfig.java – loads config.properties + dummyjson.properties  
  hooks/ Hooks.java – REST Assured setup, content-type fix, @AfterStep response logging  
  models/ UserAccount.java – POJO with 17 fields  
  runners/  
    AuthApiRunner | BrandApiRunner | ProductApiRunner | TokenAuthApiRunner |  
    UserAccountApiRunner  
    QueryParamsRunner | SortingRunner | HamcrestAssertionsRunner |  
    SchemaValidationRunner  
  services/  
    AuthService | BrandService | ProductService | UserAccountService  
    TokenAuthService | DummyJsonProductService  
  steps/  
    CommonApiSteps | AuthApiSteps | BrandApiSteps | ProductApiSteps  
    UserAccountApiSteps | TokenAuthApiSteps | DummyJsonProductsApiSteps  
src/test/resources/  
  config.properties | dummyjson.properties | cucumber.properties  
  features/  
    auth_api | brand_api | product_api | token_auth_api | user_account_api  
    query_params | sorting | hamcrest_assertions | schema_validation  
  schemas/ dummyjson-product.json | dummyjson-products-list.json
```

Test Suites

FEATURE	SCENARIOS	API	COVERAGE
product_api	5	AutomationExercise	GET list, POST 405, search, missing param 400
brand_api	2	AutomationExercise	GET list, PUT 405
auth_api	4	AutomationExercise	Valid login, missing email, DELETE 405, invalid creds
user_account_api	5	AutomationExercise	Create, read, update, delete, get after delete
token_auth_api	6	DummyJSON	JWT login, Bearer access, refresh, invalid creds, no token, invalid token
query_params	9	DummyJSON	Pagination (limit/skip), field select, category filter, search, Scenario Outline
sorting	3	DummyJSON	Sort by price asc/desc, rating desc — Scenario Outline
hamcrest_assertions	2	DummyJSON	Chained body assertions via DataTable, collection-level everyItem matchers
schema_validation	3	DummyJSON	JSON Schema for single product & list, ResponseSpec, response time

Advanced Techniques

Data-Table-Driven Assertions

REUSABLE

One step definition drives all assertions from a Cucumber DataTable. Supports 10 matchers.

```
# Feature file
And the response should match:
```

JSON Schema Validation CONTRACT

Validates API responses against JSON Schema files in `src/test/resources/schemas/`.

```
# Feature file
And the response should match the
"dummyjson-product" JSON schema
```

field	matcher	value
id	notNull	
title	notEmpty	
price	greaterThan	0
rating	between	0,5
tags	notEmptyList	

```
# Matchers available:
# notNull, notEmpty, notEmptyList,
# greaterThan, between, equalTo,
# equalToInt, listSize,
# everyItemNotNull, everyItemNull,
# everyItemEquals
```

```
# Step definition
CommonApiSteps.getResponse().then()
    .body(matchesJsonSchemaInClasspath(
        "schemas/" + schemaName + ".json"
    ));
```

```
# Dependency
io.rest-assured:json-schema-validator
```

Response Spec DRY

Reusable response specification for standard DummyJSON endpoints.

```
ResponseSpecification SPEC =
    new ResponseSpecBuilder()
        .expectStatusCode(200)
        .expectContentType("application/json")
        .expectBody("products", notNullValue())
        .expectBody("total", greaterThan(0))
        .build();
```

```
// Usage in step
CommonApiSteps.getResponse().then()
    .spec(SPEC);
```

Scenario Outline DATA-DRIVEN

Same test structure, multiple data combinations via Examples table.

Scenario Outline: GET products sorted by <field> in <order> order
When user fetches products sorted by "<field>" in "<order>" order
Then the HTTP status code should be 200
And the products should be sorted by "<field>" in "<order>" order

Examples:

field	order
price	asc
price	desc
rating	desc

Key Code Patterns

Content-Type Fix WORKAROUND

AutomationExercise returns JSON with `text/html` content type. REST Assured tries XML parser.

```
// In Hooks.java @Before
RestAssured.defaultParser = Parser.JSON;
RestAssured.registerParser(
    "text/html", Parser.JSON);

// Response filter override
((RestAssuredResponseImpl) response)
    .setContentType("application/json")
```

JWT Token Flow DUMMYJSON

Login returns `accessToken` + `refreshToken`.
Use Bearer header for protected endpoints.

```
// Login → extract tokens
accessToken = response.jsonPath()
    .getString("accessToken");
refreshToken = response.jsonPath()
    .getString("refreshToken");

// Use in request
.header("Authorization",
    "Bearer " + accessToken)
```

Response Logging REPORT

@AfterStep attaches API response body to Cucumber HTML reports.

```
@AfterStep
public void logResponseToReport(
    Scenario scenario) {
    if (CommonApiSteps.needsLogging())
        Response r = CommonApiSteps
            .getResponse();
        String out = "HTTP "
            + r.statusCode() + "\n"
            + r.asPrettyString();
        scenario.attach(out.getBytes(),
            "text/plain", "API Response");
        CommonApiSteps.markLogged();
    }
}
```

Multi-Config Properties CONFIG

Separate config files per API. No hardcoded URLs in service classes.

```
// ApiConfig.java
public static String baseUrl() {
    return props.getProperty("base.url");
}

public static String dummyJsonBaseUrl() {
    return dummyJsonProps
        .getProperty("base.url");
}

// config.properties
base.url=https://automationexercise.com

// dummyjson.properties
base.url=https://dummyjson.com
```

Run Commands

Run all tests

```
mvn clean verify
```

Run a specific suite

```
mvn test -DtestSuite=ProductApiRunner
mvn test -DtestSuite=BrandApiRunner
mvn test -DtestSuite=AuthApiRunner
mvn test -DtestSuite=UserAccountApiRunner
mvn test -DtestSuite=TokenAuthApiRunner
mvn test -DtestSuite=QueryParamsRunner
mvn test -DtestSuite=SortingRunner
mvn test -DtestSuite=HamcrestAssertionsRunner
mvn test -DtestSuite=SchemaValidationRunner
```

Jenkins parameterized build

```
Goal: clean verify -DtestSuite=${testSuite}
Parameter: Choice → testSuite
Values: *Runner | ProductApiRunner | BrandApiRunner | AuthApiRunner
        UserAccountApiRunner | TokenAuthApiRunner
        QueryParamsRunner | SortingRunner
        HamcrestAssertionsRunner | SchemaValidationRunner
```

Tech Stack

REST Assured	5.5.1
JSON Schema Validator	5.5.1
Cucumber	7.34.9
JUnit	4.13.2
Jackson	2.18.3

APIs Under Test

AutomationExercise

<https://automationexercise.com/api>

Products, Brands, Auth, User Accounts

DummyJSON

<https://dummyjson.com>

JWT auth, Products (query params, sorting, pagination, search, categories)

Java	11
Maven Surefire	3.5.6

GitHub & Jenkins

Repo	github.com/jgupta-git/restapi-automation-java-restassured-junit
Branch	main
Jenkins Job	restapi-automation-run (Maven project, parameterized)
Maven Goal	clean verify -DtestSuite=\${testSuite}
Post-Build	Publish HTML reports → target/cucumber-html-reports → overview-features.html